



Alexandria University
Alexandria Engineering Journal

www.elsevier.com/locate/aej
www.sciencedirect.com



On the dissipativity property of negative imaginary systems



Mohamed. A. Mabrok^{a,*}, Maryam Ahmed Alyami^b, Emad E. Mahmoud^{c,d}

^a Department of Mathematics and Physics, School of Engineering at the Australian College of Kuwait, Kuwait

^b Department of Mathematics, Faculty of Sciences, University of Jeddah, Jeddah, Saudi Arabia

^c Department of Mathematics and Statistics, College of Science, Taif University, P.O. Box 11099, Taif 21944, Saudi Arabia

^d Department of Mathematics, Faculty of Science, Sohag University, Sohag 82524, Egypt

Received 2 September 2020; revised 6 October 2020; accepted 23 October 2020

Available online 7 November 2020

KEYWORDS

Dissipativity;
 Passivity;
 Negative imaginary systems

Abstract This paper extends the concept of dissipativity to a new class of dynamical systems known as negative imaginary systems. The paper also introduce a new definition for nonlinear negative imaginary systems to extend the existent definition. Two different quadratic dynamic supply rates are introduced to allow for extending the dissipativity concept to cover the negative imaginary systems. One of which is a differential operator and the other is an integral operator. Both supply rates are used in order to formulate the class of negative imaginary systems as dissipative systems. This extension allows for wider class of dynamical systems to be considered in the negative imaginary framework. We also show how the new definition extends the negative imaginary system to analyze a class of higher order evolutionary dynamics. In particular, we show that the second order replicator dynamics satisfy the negative imaginary property and hence we can conclude convergence with certain class of games. Also, a nonlinear negative imaginary lemma based on the above definitions is derived.

© 2020 The Authors. Published by Elsevier B.V. on behalf of Faculty of Engineering, Alexandria University. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

The general theory of *dissipativity* was initially formulated and introduced by Willems [1,2] to characterize a large and important class of dynamical systems based on the behaviour of their

inputs and outputs. Willems defined dissipative systems as these systems that the increase in their internal stored energy is bounded from above by the quantity of energy that is supplied to the system from the environment throughout a scalar valued function known as *supply rate* $s(u, y)$.

Two important quantities identify the dissipativity property, the *storage function* $S_f(x)$ and the supply rate $s(u, y)$. In particular, the supply rate plays an essential role in defining the stability property of the dissipative systems and their interconnections. For instance, the global stability for interconnected subsystems depends on the compatibility of their supply rates, which means that these subsystems must have

* Corresponding author at: Department of Mathematics and Physics, School of Engineering at the Australian College of Kuwait, Kuwait. E-mail addresses: m.a.mabrok@gmail.com (M.A. Mabrok), maalyami@uj.edu.sa (M.A. Alyami), e.mahmoud@tu.edu.sa (E.E. Mahmoud).

Peer review under responsibility of Faculty of Engineering, Alexandria University.

the same form of supply rate. In fact, the subclass of dissipative systems such as passive systems are clustered according to their supply rate. The storage function then validates the existence of the dissipative property according to particular supply rate.

A quadratic supply rate described by a matrix triple (Q, S, R) is frequently used in the control literature. This was introduced in [3] to generalize many of the existed dissipativity results. Based on this unified supply rate, a matrix form-like condition is given to enforce the input-output stability and the Lyapunov stability conditions. Different choices of the matrices (Q, S, R) will lead to different classes of systems.

Dissipativity theory is fundamentally connected to the well known Kalman Yakubovich Popov Lemma or the KYP lemma [4, 5], which is an important result in linear control theory due to its applications in optimal control design, adaptive and learning control, absolute stability, filtering, and stochastic control. The KYP lemma characterizes the class of systems known as *positive real systems* and establishes relations between frequency domain conditions of the system the time domain condition for same class of systems. Positive real systems are in fact dissipative systems with the quadratic supply rate $(0, I, 0)$.

In spite of the generality of quadratic supply rates (Q, S, R) , there are classes of dissipative systems that are not captured due to the constant nature of the three matrices (Q, S, R) . Several attempts for dynamical supply rates have been made to capture more general dissipative systems, see for instance [6, 7]. The quadratic differential forms can be used to generalize the supply rate to capture more information on system dynamics compared with the use of conventional constant supply rates.

In this paper, we show that a class of systems known as *negative imaginary systems* can be formulated as dissipative systems if we allow the supply rate to be dynamic. In particular, two dynamical supply rates are introduced. One of which is a differential operator and the other is an integral operator. Both supply can bees are used in order to formulate the class of negative imaginary as dissipative systems. Introducing integral and differential supply rates have several practical development in the negative imaginary system theory, in particular, in studying the nonlinear negative imaginary system. For instance, systems such as robotic manipulator and quad-rotors are nonlinear systems that satisfy the negative imaginary property and they can be characterized using a suitable differential supply rates.

The theory of NI systems was developed for linear systems in [8–11] and a generalized definition was introduced in [12, 13]. They showed that negative imaginary systems theory is a suitable for modeling dynamical systems that have flexible structure components. Basically, the negative imaginary theory can be viewed as a complementary to the well known passivity theory. Negative imaginary property appears in the systems where a combination of position sensors and force actuators are used in a collocated setup.

In dynamical systems with single-input and single-output (SISO), one can define the NI property by taking into account the imaginary part of the frequency response of the system $H(j\omega)$, where the condition

$$j(H(j\omega) - H(j\omega)^*) \geq 0,$$

for all $\omega \in (0, \infty)$, must be satisfied.

The stability of the closed loop interconnected systems that satisfy the negative imaginary property has been investigated and developed in [8, 9]. In these references, the necessity and sufficient condition for the closed loop stability is given as a De-gain condition as follows;

$$\Upsilon_{\max}(H(0)\bar{H}(0)) < 1. \quad (1)$$

Here, $\Upsilon_{\max}(\cdot)$ means the maximum eigenvalue and $\bar{H}$ is the complex conjugate transpose of H .

In [14], a form of static-output controller of feedback interconnection of a negative imaginary systems was developed. Also, the discrete-time version that capture the main results in the negative imaginary system theory in discrete time domain was introduced and developed in [15, 16]. In these papers, the authors derived relationship between NI notion and discrete-time positive real notion. Also, in [17] similar discrete-time result was considered.

Our paper is organized in the following flow: In Section 2 we present a formal existing definition of dissipative systems. Section 3 presents a motivational example to motivate the topic discussed in this paper. Section 4 introduces the variation of the definitions of the negative imaginary systems and the discuss their connections to dissipativity theory.

2. Dissipative Systems

Dissipativity was formulated and introduced by Willems [1, 2]. In Dissipativity framework, dynamical systems are characterized based on their inputs and outputs through a scalar valued function known as the supply rate $S(u, y)$.

In order to present the formal definition of dissipative, we consider the following nonlinear system with state space representation as follows;

$$\dot{x} = F(x, u) \quad F(0, 0) = 0, \quad (2)$$

$$y = T(x, u) \quad T(0, 0) = 0. \quad (3)$$

Here, the vector $x(t) \in \mathbb{R}^n$ represents the state of the dynamics, $u(t) \in \mathbb{R}^m$ represents the input of the system, and $y(t) \in \mathbb{R}^p$ represents the output of the system. The function $F: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ satisfies the Lipschitz continuity condition and represents the vector field whose components are the dynamical rules governing the nature of the dynamical variables. The function $T: \mathbb{R}^n \rightarrow \mathbb{R}^p$ is a type C^1 function.

Definition 1. The dynamical system defined in (2) and (3) is said to be dissipative with respect to the supply rate $s(u(t), y(t))$ if there exist a function $S_f(\cdot)$, named a storage function where, $S_f(x(t)) \geq 0, S_f(0) = 0$, where,

$$S_f(x(t)) \leq S_f(x_0) + \int_0^t s(u(\tau)y(\tau))d\tau, \quad (4)$$

for all input signals $u(\cdot)$ and every $\tau \geq 0$ whenever $x(t)$ exists. Here, $S_f(x_0)$ represents the storage function at the initial states x_0 .

The supply rate $s(u(t), y(t))$ in the above definition plays a key role of the characterization in the dissipativity property. Accordingly, the choice of the type of the supply rate $s(u(t), y(t))$ defines different classes of systems.

One of the most common and general form of supply rate used in the control theory literature is the (Q, S, R) supply rate, which is defined as follows;

$$s(u, y) = y^T Q y + 2y^T S u + u^T R u \quad (5)$$

$$= \begin{bmatrix} y^T & u^T \end{bmatrix} \begin{bmatrix} Q & S \\ S^T & R \end{bmatrix} \begin{bmatrix} y \\ u \end{bmatrix}.$$

Here, the matrices Q, R and S are constant matrices.

Several classes of systems can be generated from the above generalized supply rate. For instance,

- **Passivity:** If the supply-rate given in (5) is reduced to

$$s(u(t), y(t)) = y^T u,$$

then, the dissipativity property is reduced to the very well known passivity property.

- **Output strict passivity:** If the supply-rate given in (5) is reduced to

$$s(u(t), y(t)) = y^T u - \epsilon |y(t)|^2,$$

with $\epsilon > 0$, then the dissipativity property reduces to the output strictly output passivity property.

- **Finite gain:** If the supply-rate given in (5) is reduced to

$$s(u(t), y(t)) = \epsilon |u(t)|^2 - |y(t)|^2,$$

with $\epsilon > 0$, then the dissipativity property reduces to the finite gain property.

In the next section, a motivational example will be presented in order to demonstrate that above dissipativity definition do not include an important class of dynamical systems.

3. Motivational example

Here, we present an important example to show a dissipative system which cannot be captured by a constant supply rate (Fig. 1).

Let us consider a simple spring-damper-mass system with single mass and damper as shown in Fig. 2.

The differential equation of motion that corresponds to the spring-damper system can be obtained from Newton's second law as follows:

$$- \beta \dot{x} - \alpha \ddot{x} + F_e = m \ddot{x}. \quad (6)$$

This can be written as follows;

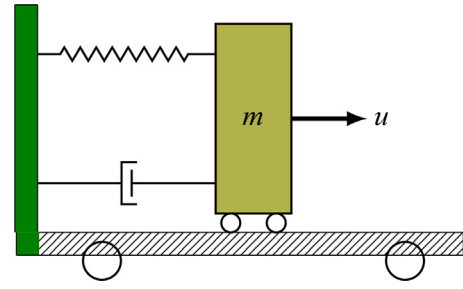


Fig. 2 The spring-damper-mass system that represent a simple negative imaginary system.

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = u, \quad (7)$$

where, $\omega_n = \sqrt{\frac{\beta}{m}}$, $\zeta = \frac{\alpha}{2m\omega_n}$, $u = \frac{F_e}{m}$. The $x(t)$ represents the position, at time t , of mass m , which is attached to the spring, the constant α is the viscous damping coefficient, β is the spring constant, and u is the applied input force.

The corresponding transfer function from the input force $u(t)$ to the output displacement $y(t) = x(t)$ of the above system in the frequency domain is given as follows;

$$H(s) = \frac{1}{ms^2 + cs + k} \quad (8)$$

Since the relative degree between the numerator and the denominator is two. Therefore, the transfer function (8) is not a passive system with respect to the quadratic supply (Q, S, R) . Although, the system itself is dissipative from the physical point of view due to the damping. However, if we allow a dynamical supply rate such as $s(u, y) = u\dot{y}$, then, the system is dissipative.

It is important to realize that the physical representation of the systems that satisfy the negative imaginary property is different from the systems that satisfy the positive real property. A linear time invariant (LTI) dynamical system that has transfer function matrix with relative-degree (the difference between the order of the denominator polynomial or the number of poles and the order of numerator polynomial or the number of zeros) more than one do not satisfy the positive real property. However, in the case of negative imaginary transfer function, the relative-degree between the number of poles and the number of zeros can be up to two. This fact can clearly appear when a dynamical system consists of force actuators when they are combined with position sensors. This combination of

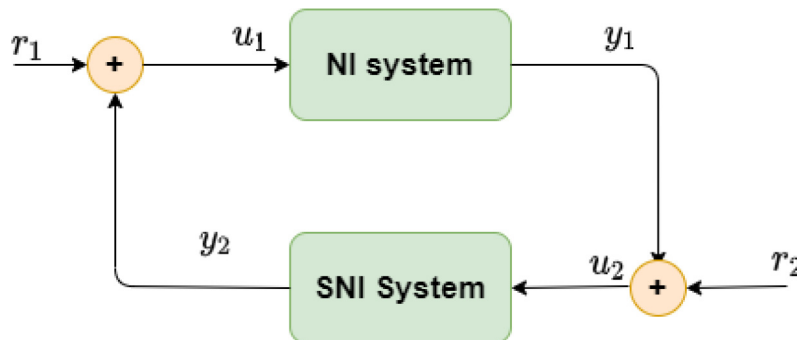


Fig. 1 Negative imaginary positive feedback systems.

actuators and position sensors are commonly found in robotics systems for instance. In [18], several physical interpretations and realization of negative imaginary systems are discussed.

In the next section, we will present two possible dynamical supply rates to capture such systems and connect the negative imaginary property with the dissipativity property.

4. Dissipative negative imaginary systems

This section introduces two supply rates that allow the class of negative imaginary systems to be included into the class of dissipative systems.

4.1. Linear negative imaginary systems

In this subsection, we briefly, recite the formal definition of linear-time-invariant (LTI) negative imaginary systems as introduced in [12].

Generally, an LTI systems has the state-space realization:

$$\dot{x}(t) = Ax(t) + Bu(t), \quad (9)$$

$$y(t) = Cx(t) + Du(t), \quad (10)$$

where the matrix A is a square matrix with $A \in \mathbb{R}^{n \times n}$, the matrix $B \in \mathbb{R}^{n \times m}$, the matrix $C \in \mathbb{R}^{m \times n}$, and the matrix $D \in \mathbb{R}^{m \times m}$. In addition, a transfer function $H(s)$ is defined as strictly proper if $H(\infty) = 0$ i.e., $D = 0$.

Now, we present the NI definition as defined in [12];

Definition 2. [12,19] For the dynamical systems with same number of inputs and outputs, the transfer function $H(s)$ is said to be NI if the following items holds:

1. The transfer function $H(s)$ has no pole in the right half of the complex plane, i.e., $\text{Re}[s] > 0$.
2. For all positive frequency, $\omega > 0$, such that $s = j\omega$ is not one of the poles of $H(s)$, then

$$j(H(j\omega) - H(j\omega)^*) \geq 0. \quad (11)$$

3. If $s = j\omega_0$ with $\omega_0 > 0$ is one of the poles of $H(s)$, then it must be simple pole, where, the residue matrix $\Omega = \lim_{s \rightarrow j\omega_0} (s - j\omega_0)jH(s)$ is positive semidefinite and Hermitian matrix.

4. If $s = 0$ is a pole of $H(s)$, then $\lim_{s \rightarrow 0} s^i H(s) = 0$ for all $i \geq 3$ and $\lim_{s \rightarrow 0} s^2 H(s)$ is Hermitian and positive semidefinite.

The following definition for the strictly negative imaginary systems (SNI) as presented in [20].

Definition 3. [20] The transfer function $H(s)$ is SNI if the following these two conditions hold:

1. The transfer function $H(s)$ has no pole in $\text{Re}[s] \geq 0$.
2. For all $\omega > 0$, $j(H(j\omega) - H(j\omega)^*) > 0$.

Here, we present an NI lemma.

Lemma 1. [8] Let

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right]$$

defining the system (9) and (10) be a minimal state space representation of the transfer function matrix $H(s)$. Then, $H(s)$ is negative imaginary if and only if the matrix $D = D^T$ and there exist matrices $\Pi = \Pi^T \geq 0$, $\Gamma \in \mathbb{R}^{m \times m}$, and $L \in \mathbb{R}^{m \times n}$ such that the following linear matrix inequality (LMI) is satisfied:

$$\begin{bmatrix} \Pi A + A^T \Pi & \Pi B - A^T C^T \\ B^T \Pi - C A & -(CB + B^T C^T) \end{bmatrix} = \begin{bmatrix} -L^T L & -L^T \Gamma \\ -\Gamma^T L & -\Gamma^T \Gamma \end{bmatrix} \leq 0. \quad (12)$$

4.2. Dissipative negative imaginary systems

Negative imaginary systems can be recovered from dissipative systems if one allows for a dynamic supply rate. In fact, dynamic supply rates have been discussed in several occasions; see for instance, [6,7].

The above definition of negative imaginary systems can be recovered from following supply rate:

Supply rate with a differential operator:

$$s_D(u, y) = \begin{bmatrix} u^T & y^T \end{bmatrix} \begin{bmatrix} 0 & S_D \\ S_D^T & 0 \end{bmatrix} \begin{bmatrix} u \\ y \end{bmatrix} = 2u^T S_D y, \quad (13)$$

where, the matrix S_D is the differential operator defined as follows;

$$S_D(y) = I \frac{d}{dt} y.$$

The above supply rate leads to the following general negative imaginary definition.

Now, for following nonlinear dynamical system;

$$\dot{x} = f(x, u); \quad (14)$$

$$y = h(x), \quad (15)$$

where the function $f: \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ satisfy the Lipschitz continuity condition and the function $h: \mathbb{R}^n \rightarrow \mathbb{R}$ is a type C^1 function.

Definition 4. The nonlinear system (14), (15) is Negative Imaginary if there exists a positive storage function $S_f: \mathbb{R}^n \rightarrow \mathbb{R}$ of a class C^1 such that

$$S_f(x(t)) \leq S_f(x(0)) + \int_0^t u(s)^T \dot{y}(s) ds, \quad (16)$$

for all $t > 0$.

Alternatively, the integral condition (16) can be represented by the differential condition

$$\dot{S}_f(x(t)) \leq u(t)^T \dot{y}(t). \quad (17)$$

The above definition corresponds the existing nonlinear negative imaginary definition in [21]. This definition was motivated by the passivity definition.

4.3. Integral supply rate

Another class of negative imaginary systems can be recovered from following supply rate:

$$s_I(u, y) = \begin{bmatrix} y^T & u^T \end{bmatrix} \begin{bmatrix} 0 & S_I \\ S_I^T & 0 \end{bmatrix} \begin{bmatrix} y \\ u \end{bmatrix} \quad (18)$$

$$= 2y^T S_I u.$$

where, the matrix S_I is the integral operator defined as follows;

$$S_I(u) = I \int_0^\tau u(\tau) d\tau. \quad (19)$$

The above supply rate defined in (19) leads to the following general definition of negative imaginary systems.

Definition 5. The nonlinear system (14), (15) is negative imaginary if there exists a positive storage function $S_f: \mathbb{R}^n \rightarrow \mathbb{R}$ of a class C^1 such that

$$S_f(x(t)) \leq S_f(x(0)) + \int_0^t y(s)^T \int_0^s u(\tau) d\tau ds, \quad (20)$$

for all $t > 0$.

Alternatively, this condition (20) can be replaced by the differential condition

$$\dot{S}_f(x(t)) \leq y(t)^T \int_0^t u(\tau) d\tau. \quad (21)$$

The above definition added a new sub-class to the negative imaginary systems, which was not included in the previous definitions. One can illustrate the above definition by considering the linear case. It is straightforward to show that if a given transfer function $H(s)$ satisfy the negative imaginary property from u to y , then the transfer function from the integration of input u to the output y satisfy the positive real property.

5. Stability analysis

One of the immediate result associated with the class of dissipative systems defined in 4 and 5 are the stability of such systems operating under no control actions. The following lemma discusses the stability of nonlinear negative imaginary systems defined in 4 and 5.

Lemma 2. If the system (14) and (15) is negative imaginary with a positive storage function $S_f(x)$ then the unforced system

$$\dot{x} = f(x, 0) \quad (22)$$

is Lyapunov stable.

Proof. Since the system (14) and (15) is negative imaginary, then, it follows that there is a positive storage function $S_{f_1}(x)$ such that

$$\dot{S}_{f_1}(x(t)) \leq y(t)^T \int_0^t u(\tau) d\tau, \quad (23)$$

or there is a positive storage function $S_{f_2}(x)$ such that

$$\dot{S}_{f_1}(x(t)) \leq u(t)^T \dot{y}(t). \quad (24)$$

This implies that for the unforced system, i.e., $u = 0$, Eqs. (23) and (24) reduces to

$$\dot{S}_{f_i}(x(t)) \leq 0 \quad i = 1, 2. \quad (25)$$

By considering $S_{f_i}(x)$ to be a Lyapunov function, this implies that the unforced system (22) is Lyapunov stable. $\square$

6. Game theory connection

The new non-linear negative imaginary definition given in 5 enlarge the class of negative imaginary systems to include new and important class of systems. For instance, the second order replicator dynamic, which is part of a class of evolutionary dynamics known as *Higher-order dynamics* cannot be satisfied the passivity property. Instead, the second order replicator dynamics satisfy the negative imaginary property using Definition 5. In the following subsection, we will briefly introduce the Higher-order dynamics and show how the second order replicator dynamics satisfy the negative imaginary property though Definition 5.

6.1. Higher-order dynamics

A standard game, in a game theoretic setup, is a static mapping from set strategies X to a set of real valued payoffs ξ ,

$$\xi = F(x).$$

Similarly, an evolutionary dynamic is a first-order dynamical mapping from the payoff trajectory ξ to the strategy trajectory X ,

$$\dot{x} = S_f(x, \xi).$$

One way to introduce higher-order dynamics into evolutionary dynamics is to auxiliary-like states to the first-order dynamics in [22,23]. Also, another way to obtain higher-dynamics is by conducting a second derivative of the regular order dynamics [24]. It is demonstrated in [22–26] that any adjustment of the standard well known first dynamics produce a variety of qualitative behaviors.

In this paper, we are following [22,23,27] to generate a new dynamics with a second ordered derivative by introducing an auxiliary-like dynamical state z into the regular. This implies the following dynamical form;

$$\begin{aligned} \dot{z} &= f(z, \xi) \quad z(0) = z_0, \\ \dot{x} &= H(z, \xi), \end{aligned}$$

where, for all $x(0)$ in the simplex and for all bounded $\xi(\cdot)$, there exists x^* and ξ^* such that if $\xi(t) \rightarrow \xi^*$, then, $x(t) \rightarrow x^*$. Here, The function $f: \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ satisfies the Lipschitz continuity condition and represents the vector field whose components are the dynamical rules governing the nature of the dynamical variables. The function $H: \mathbb{R}^n \rightarrow \mathbb{R}$ is a type C^1 function.

In the following subsections, we introduce two different forms with higher-order dynamics; one of which the replicator dynamics and the other is logit dynamics. Here, we will show that theses dynamics fits the frame of negative imaginary framework.

6.1.1. Replicator dynamics

For the replicator dynamics, the new replicator is obtained by impeding the auxiliary dynamical state $\hat{\xi}$ to the payoff trajectory. The obtained dynamics takes the following form:

$$\dot{x}_i = x_i \left(\hat{\xi}_i - \sum_j x_j \hat{\xi}_j \right) \quad (26a)$$

$$\dot{\hat{\xi}}_i = \xi_i. \quad (26b)$$

For the above dynamics, the conditions of equilibrium are $\xi_i^* = 0$, $x_i^* = \sigma_{\max}(\hat{\xi}_i^*)$, and $\hat{\xi}_i^* = 1$.

The Eq. (26b) can be written as follows;

$$\dot{\hat{\xi}}_i = \xi_i \quad (27)$$

$$\hat{\xi}_i = \int_0^t \xi_i(\tau) d\tau. \quad (28)$$

It has been shown in [28] that this new replicator dynamics is not passive. This has been shown by linearizing the obtained replicator dynamics (26), which is reduced to a two poles ant the origin and hence it is not passive.

The following proposition proves that the second order replicator defined in (26) is a negative imaginary system according to the definition given in (5).

Proposition 1. *The second order replicator dynamics defined in (26) from the error e_{x_i} to the payoff ξ is a negative imaginary system.*

Proof. Consider the following storage function,

$$S_f(e_x) = -\sum_i x_i^* \ln \frac{e_{x_i} + x_i^*}{x_i^*}, \quad (29)$$

Using $e_{x_i} = x_i - x_i^*$, the replicator dynamic Eq. (26) can be written as follows:

$$\dot{e}_{x_i} = (e_{x_i} + x_i^*) \left(\hat{\xi}_i - \sum_j (e_{x_j} + x_j^*) \hat{\xi}_j \right) \quad (30)$$

$$\hat{\xi}_i = \int_0^t \xi_i(\tau) d\tau. \quad (31)$$

Using the storage function (29), it follows that;

$$\begin{aligned} \dot{S}_f(e_x) &= -\sum_i x_i^* \frac{1}{e_{x_i} + x_i^*} \dot{e}_{x_i} \\ &= -\sum_i x_i^* \left(\hat{\xi}_i - \sum_j (e_{x_j} + x_j^*) \hat{\xi}_j \right) \\ &= -\sum_i x_i^* \hat{\xi}_i + \sum_i x_i^* \left(\sum_j e_{x_j} \hat{\xi}_j + \sum_j x_j^* \hat{\xi}_j \right) \\ &= -\sum_i x_i^* \hat{\xi}_i + \sum_j e_{x_j} \hat{\xi}_j + \sum_j x_j^* \hat{\xi}_j \\ &= \sum_j e_{x_j} \hat{\xi}_j \\ &= \sum_j e_{x_j} \int_0^t \xi_j(\tau) d\tau. \end{aligned}$$

This implies that second order replicator dynamic is negative imaginary. $\square$

Proposition 1 shows that the second order replicator dynamics defined in (26) satisfies the negative imaginary property. This implies that the feedback interconnection between the second order replicator dynamics defined in (26) and any strictly negative imaginary game should lead to a stable behaviour [29]. For instance, consider the following second order negative imaginary game:

$$\begin{aligned} \dot{z} &= Az + Bu \\ \xi &= Cz + Du, \end{aligned} \quad (32)$$

where,

$$A = \begin{bmatrix} -0.3 & 0 \\ 0 & -1.3 \end{bmatrix}, B = C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

and $D = \begin{bmatrix} -5.7333 & 0 \\ 0 & -3.1665 \end{bmatrix}.$

The feedback interconnection between the second order replicator dynamics (26) and the second order strictly negative imaginary game (32) is demonstrated in Fig. 3.

The following Fig. 3, represent the dynamics and the changes in the state trajectories of the feedback loop, in positive interconnection, between the given game (32) that satisfies the strictly negative imaginary property and our new second-order replicator dynamics (26).

6.1.2. Logit dynamics

The logit dynamics was introduced in [30] as another version of the dynamics known as best response dynamics. The second order version of the logit dynamics can be derived similarly to the replicator dynamics by impeding an auxiliary-like dynamical state into the payoff (the output of the game) trajectory as follows:

$$\dot{x}_i = -x_i + \frac{e^{\hat{\xi}_i}}{\sum_j e^{\hat{\xi}_j}} \quad (33a)$$

$$\dot{\hat{\xi}}_i = -\hat{\xi}_i + \xi_i. \quad (33b)$$

Here, x represents the state-vector, ξ represents the payoff trajectory and $\hat{\xi}$ represents the new auxiliary-like dynamical state. And therefore, the equilibrium conditions are $\hat{\xi} = \xi$ and the $x_i^* = \sigma_{\max}(\hat{\xi}_i^*)$.

Similarly to the replicator dynamics in the above section, one can conclude that the second order logit dynamic satisfy the negative imaginary property.

To demonstrate stability of the feedback interconnection of the Logit dynamics with a higher order game, consider the following second order negative imaginary game:

$$\begin{aligned} \dot{z} &= Az + Bu \\ \xi &= Cz + Du, \end{aligned} \quad (34)$$

where,

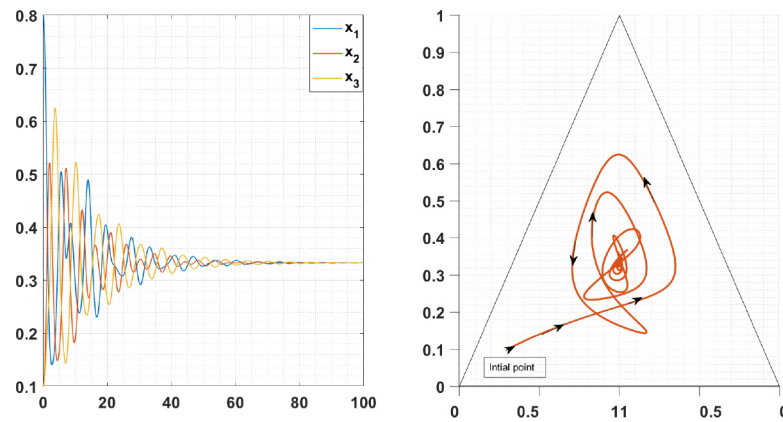


Fig. 3 On the left, the evolution of the state trajectories of the strictly negative imaginary game (32) in feedback with the new replicator dynamics (26) and in the right the same projected on the simplex.

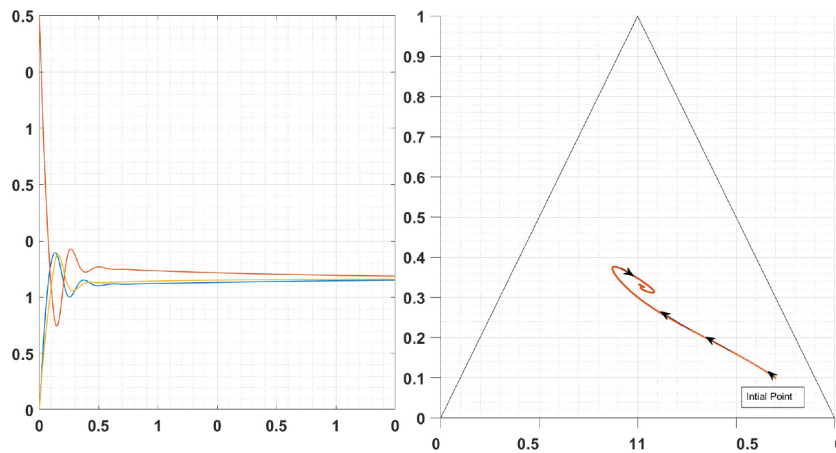


Fig. 4 On the left, the evolution of the state trajectories of the strictly negative imaginary game (34) in feedback with the logit dynamics (33a) and (33b) and in the right the same projected on the simplex.

$$A = \begin{bmatrix} -0.1 & 0 \\ 0 & -0.3 \end{bmatrix}, B = C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

and $D = \begin{bmatrix} -10.3 & 0 \\ 0 & -4.1665 \end{bmatrix}.$

The feedback interconnection between the second order logit dynamics (33a) and (33b) and the second order strictly negative imaginary game (34) is demonstrated in Fig. 4.

7. Conclusion

This paper established the relationship between the dissipativity property and the class of negative imaginary systems. Two supply rates involving integral and differential operators are introduced to capture the negative imaginary systems. The paper also introduced a new definition for nonlinear negative imaginary systems to extend the existent definition. The paper also extended the notion of negative imaginary to game theory to analyze a class of higher order evolutionary dynamics. In particular, we demonstrated that the new replicator dynamics (the second-order) and the new logit dynamics (second order) satisfies the negative imaginary property.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

Taif University Researchers Supporting Project number (TURSP-2020/20), Taif University, Taif, Saudi Arabia.

References

- [1] J.C. Willems, Dissipative dynamical systems part I: general theory, *Arch. Ration. Mech. Anal.* 45 (1972) 321–351.
- [2] J.C. Willems, Dissipative dynamical systems part ii: linear systems with quadratic supply rates, *Arch. Ration. Mech. Anal.* 45 (1972) 352–393.
- [3] P. Moylan, D. Hill, Stability criteria for large-scale systems, *IEEE Trans. Autom. Control* 23 (1978) 143–149.
- [4] V. Jakubovic, The solution of certain matrix inequalities in automatic control, *Soviet Math.* 3 (1962) 504.

- [5] R.E. Kalman, Lyapunov functions for the problem of Lur'e in automatic control, *Proc. Natl. Acad. Sci.* 49 (1963) 201–205.
- [6] J.C. Willems, K. Takaba, Dissipativity and stability of interconnections, *Int. J. Robust Nonlinear Cont. IFAC-Affiliated J.* 17 (2007) 563–586.
- [7] M.J. Tippet, J. Bao, Dissipativity based analysis using dynamic supply rates, *IFAC Proc. Vol.* 44 (2011) 1319–1325.
- [8] A. Lanzon, I.R. Petersen, Stability robustness of a feedback interconnection of systems with negative imaginary frequency response, *IEEE Trans. Autom. Control* 53 (2008) 1042–1046.
- [9] I.R. Petersen, A. Lanzon, Feedback control of negative imaginary systems, *IEEE Contr. Syst. Mag.* 30 (2010) 54–72.
- [10] V.P. Tran, F. Santoso, M.A. Garratt, I.R. Petersen, Adaptive second-order strictly negative imaginary controllers based on the interval type-2 fuzzy self-tuning systems for a hovering quadrotor with uncertainties, *IEEE/ASME Trans. Mechatron.* 25 (2020) 11–20.
- [11] V.P. Tran, M. Garratt, I.R. Petersen, Switching time-invariant formation control of a collaborative multi-agent system using negative imaginary systems theory, *Contr. Eng. Practice* 95 (2020) 104245.
- [12] M. Mabrok, A. Kallapur, I. Petersen, A. Lanzon, Generalizing negative imaginary systems theory to include free body dynamics: Control of highly resonant structures with free body motion, *IEEE Trans. Autom. Control* 59 (2014) 2692–2707.
- [13] M. Mabrok, Controller synthesis for negative imaginary systems using nonlinear optimisation and h_2 performance measure, *Int. J. Control* (2019) 1–9.
- [14] J. Xiong, J. Lam, I.R. Petersen, Output feedback negative imaginary synthesis under structural constraints, *Automatica* 71 (2016) 222–228.
- [15] A. Ferrante, A. Lanzon, L. Ntogramatzidis, Foundations of negative imaginary systems theory and relations with positive real systems, *arXiv preprint arXiv:1412.5709* (2014).
- [16] A. Ferrante, A. Lanzon, L. Ntogramatzidis, Discrete-time negative imaginary systems, *Automatica* 79 (2017) 1–10.
- [17] M. Liu, J. Xiong, Properties and stability analysis of discrete-time negative imaginary systems, *Automatica* 83 (2017) 58–64.
- [18] I.R. Petersen, Physical interpretations of negative imaginary systems theory, in: 2015 10th Asian Control Conference (ASCC), pp. 1–6.
- [19] M. Mabrok, A. Kallapur, I. Petersen, D. Schutte, T. Boyson, A. Lanzon, Locking a three-mirror optical cavity: A negative imaginary systems approach, in: Control Conference (AUCC), 2012 2nd Australian, pp. 476–480.
- [20] J. Xiong, I.R. Petersen, A. Lanzon, A negative imaginary lemma and the stability of interconnections of linear negative imaginary systems, *IEEE Trans. Autom. Control* 55 (2010) 2342–2347.
- [21] A.G. Ghallab, M.A. Mabrok, I.R. Petersen, Extending negative imaginary systems theory to nonlinear systems, in: 2018 IEEE Conference on Decision and Control (CDC), IEEE, pp. 2348–2353.
- [22] J.S. Shamma, G. Arslan, Dynamic fictitious play, dynamic gradient play, and distributed convergence to nash equilibria, *IEEE Trans. Autom. Control* 50 (2005) 312–327.
- [23] G. Arslan, J.S. Shamma, Anticipatory learning in general evolutionary games, in: 45th IEEE Conference on Decision and Control, San Diego, CA, pp. 6289–6294.
- [24] R. Laraki, P. Mertikopoulos, Higher order game dynamics, *J. Econ. Theory* 148 (2013) 2666–2695.
- [25] S. Hart, A. Mas-Colell, Uncoupled dynamics do not lead to Nash equilibrium, *Am. Econ. Rev.* 93 (2003) 1830–1836.
- [26] S. Sato, E. Akiyama, J.D. Farmer, Chaos in learning a simple two person game, *Proc. Nat. Acad. Sci.* 99 (2002) 4748–4751.
- [27] M.J. Fox, J.S. Shamma, Population games, stable games, and passivity, *Games* 4 (2013) 561–583.
- [28] M. Mabrok, J.S. Shamma, Passivity analysis of higher order evolutionary dynamics and population games, in: Decision and Control (CDC), 2016 IEEE 55th Conference on, IEEE, pp. 6129–6134.
- [29] M.A. Mabrok, A.G. Kallapur, I.R. Petersen, A. Lanzon, A stability result on the feedback interconnection of negative imaginary systems with poles at the origin, in: Proceedings of the 2012 Australian Control Conference, Sydney, Australia.
- [30] D. McFadden, Conditional logit analysis of qualitative choice behavior, *Front. Econom.* (1974) 105–142.